

CURRICULUM VITAE



Dr. MICHAEL MAHESH K, B.E, M.E., Ph.D.

Professor, Department of ECE, Saveetha Engineering College, Chennai

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[Clarivate](#) | [Google Scholar](#) | [Scopus](#) | [michaelmahesh.github.io](#)

CAREER OBJECTIVE

To contribute as a Professor and researcher toward advancing Artificial Intelligence, Deep Learning and Computer Vision through quality teaching, doctoral supervision, impactful research and institutional development, while continuing to sharpen my expertise in emerging areas such as Vision-Language Models and Generative AI for Healthcare.

ACADEMIC QUALIFICATION

Sl. No.	Name of the Degree with Specialization	College Name	Year of Passing	University
1	Ph.D. - Information & Communication Engineering	Anna University, Chennai.	2021	Anna University
2	M.E - Computer & Communication Engineering	Sri Sairam Engineering College, Chennai.	2011	Anna University
3	B.E - Electronics & Communication Engineering	St. Xavier's Catholic College of Engineering, Nagercoil.	2009	Anna University

RESEARCH SUPERVISOR RECOGNITION

Sl. No.	Reference No	Department	Recognized University
1	4040049	Electronics and Communication Engineering	Anna University, Chennai

PH.D. RESEARCH SUPERVISION (CURRENTLY GUIDING)

Sl. No.	Scholar Name	Registration Year	Current Status
1	Santhosh S	2024	Confirmation Completed
2	Creesy R	2024	Confirmation Completed
3	Sudha R	2024	Course Work
4	Indumathy R	2024	Course Work

Sl. No.	Scholar Name	Registration Year	Current Status
5	Anushia R M	2025	Course Work
6	Jaashika K M	2025	Course Work

TEACHING EXPERIENCE: 15+ YEARS

Sl. No.	Institution	Designation	From	To	Experience (YY-MM)
1	Saveetha Engineering College, Chennai	Professor ECE	Jun 2026	Present	-
2	Saveetha Engineering College, Chennai	Assoc. Prof ECE	Dec 2023	May 2026	02.06
3	SRM Institute of Science & Technology, Ramapuram	Asst. Prof ECE	Jul 2023	Nov 2023	00.05
4	St. Joseph College of Engineering	Assoc. Prof ECE	Apr 2023	Jul 2023	00.03
5	St. Joseph College of Engineering	Asst. Prof ECE	Jul 2011	Apr 2023	11.09

Total teaching experience: 15+ years | *Experience after Ph.D.: 5+ years*

RESEARCH SPECIALIZATION (MAJOR SCIENTIFIC FIELD OF INTEREST)

- Artificial Intelligence, Deep Learning, Machine Learning, Computer Vision
- Medical Image Processing, Medical AI, Healthcare AI
- Advanced Driver Assistance Systems (ADAS), Intelligent Transportation Systems
- Evolutionary & Metaheuristic Optimization, Intelligent Systems
- Vision Transformers, Foundation Models, Multimodal AI, Vision-Language Models, Generative AI for Healthcare

RESEARCH EXPERIENCE

- Proposed MICH-Net with African Fire Hawk Optimization for steering angle prediction in ADAS.
- Developed PMFHNet for road anomaly object detection under low-light conditions.
- Proposed RBCO-based DCNN to classify the intersection point in the Road Segment.
- Proposed TimeRide-NN classifier algorithm for anomaly detection in surveillance video.
- Developed a new DeepJoint Segmentation algorithm for MRI brain images.
- Proposed FJO, FJWO based DCNN for brain tumor severity-level classification.

RESEARCH GRANT

Sl. No.	Academic Year	Title of the Project	Funding Agency	Amount Sanctioned (Rs.)
1	2022-23	Autonomous Mobile Robot Vehicle for Medical Delivery to Hospitalized Patients	TNSCST	10,000/- (Applied)
2	2021-22	Intelligent Covid-19 Pandemic Bus Service with Safety Measure (ECV-404)	TNSCST	7,500/- (Sanctioned)
3	2021-22	Teaching STEMM to Rural School Students and Teachers Through Conceptual Frameworks and Series of Lectures	DST	3,50,000/- (Applied)

DETAILS OF JOURNAL PUBLISHED

Sl. No.	Title	Name of the Journal / Type	Impact Factor / Year / ISSN
1	MICH-Net: A Novel Deep Learning Architecture With African Fire Hawk Optimization for Steering Angle Prediction in an ADAS	IEEE Transactions on Intelligent Transportation Systems SCI Q1	SJR: 2.608 2025
2	HY-LSTM: A New Time Series Deep Learning Architecture for Estimation of Pedestrian Time to Cross in ADAS	Journal of Visual Communication and Image Representation Elsevier SCI Q1	SJR: 0.573 2023 ISSN: 1047-3203
3	DeepDrive: A Braking Decision-Making Approach Using Optimized GAN and Deep CNN for ADAS	Engineering Applications of Artificial Intelligence Elsevier SCI Q1	SJR: 1.782 2023 ISSN: 0952-1976
4	Evolutionary Intelligence for Brain Tumor Recognition from MRI Images: A Critical Study and Review	Evolutionary Intelligence Springer ESCI Q1	SJR: 0.700 2018 ISSN: 1864-5909
5	PMFHNet: Parallel Maxout Feed Harmonic Network for Road Anomaly Object Detection in ADAS Under Low Light Conditions	The Imaging Science Journal Q2	SJR: 0.321 2025/2026
6	Elk Skill Optimizer Enabled Joint Channel Access and Energy Management Algorithm for Dynamic Channel Allocation in WSN	International Journal of Communication Systems Q2	SJR: 0.419 2026
7	Decentralized Control Design for UAV Swarms Communication	Discover Applied Sciences Q2	SJR: 0.573 2025

Sl. No.	Title	Name of the Journal / Type	Impact Factor / Year / ISSN
8	RBorderNet: Rider Border Collie Optimization-based DCNN for Road Scene Segmentation and Road Intersection Classification	Digital Signal Processing Elsevier SCI Q2	SJR: 0.704 2022 ISSN: 1051-2004
9	Detection and Localization of Abnormalities in Surveillance Video Using Timerider-Based Neural Network	The Computer Journal SCI Q2	SJR: 0.441 2021 ISSN: 0010-4620
10	DeepJoint Segmentation for the Classification of Severity-Levels of Glioma Tumour Using Multimodal MRI Images	IET Image Processing SCI Q2	SJR: 0.569 2020 ISSN: 1751-9667
11	Multiclassifier for Severity-Level Categorization of Glioma Tumors Using Multimodal MRI Brain Images	International Journal of Imaging Systems and Technology (IMA) SCI Q2	SJR: 0.601 2020 ISSN: 0899-9457
12	Multi-Modal Lung Cancer Detection Using Pyramidal Cascade Neuro-Fuzzy Fractional Network	Cancer Investigation Q3	SJR: 0.495 2025
13	Deep Learning-Based Energy Prediction and Tangent Search Remora Optimization-Based Secure Multi-Path Data Communication Mechanism in WSN	Network: Computation in Neural Systems Q4	SJR: 0.290 2025
14	Workflow Scheduling Using Improved Moth Swarm Optimization Algorithm in Cloud Computing	International Journal on Multimedia Research	2020
15	Deep Kronecker NasNet-based AB-M Tree for Storage and Efficient Retrieval Model using Blockchain	Iranian Journal of Science and Technology, Transactions of Electrical Engineering Q3	2026
16	Optimized Parasitic Egg Detection and Classification Using Transfer Learning with Coot Tangent Search using microscopic images	Sādhanā — Academy Proceedings in Engineering Sciences Q2	2026
17	Deep Learning Technique for rice leaf disease classification and severity level identification via Hybrid ResConvolutional Neural Network	Scientific Reports Q1	2026

Sl. No.	Title	Conference / Year	DOI
1	Detecting Early-Stage Stress Levels Using EEG and Heart Rate Monitoring with Enhanced Sparse Fire Hawk Graph Probabilistic Spiking Attention Neural Network	3rd Intl. Conf. on Intelligent Data Communication Technologies and IoT, 2025	10.1109/IDCIOT64235.2025.10915029
2	Improving FANET Communication with a QoS Optimized Location-Aided Routing Protocol	5th Intl. Conf. on Smart Electronics and Communication, 2024	10.1109/ICOSEC61587.2024.10722211
3	Next-Generation Wearable Antenna: Optimized Multi-Band Performance for Advanced Wireless Systems and Applications	5th Intl. Conf. on Circuits, Control, Communication and Computing, 2024	10.1109/I4C62240.2024.10748486
4	Design and Implementation of a Bio-Inspired Robot Arm: Machine Learning, Robot Vision	Intl. Conf. on New Frontiers in Communication, Automation, Management and Security, 2023	10.1109/ICCAMS60113.2023.10526171

PATENT PUBLICATION

Sl. No.	Application Number	Title	Application Date / Published	Status
1	IN 201941045486	PPG Signal-Based Biometric Recognition for Smartwatch Locking System	08.11.2019	Under Examination
2	IN 202341000695	Artificial Intelligence based Smart Healthcare Monitoring System to predict and diagnose Lungs cancer and breast cancer using WSN, Machine Learning, Big Data and Image Processing	04.01.2023	Under Examination
3	463340-001	IoT-Based Device to Measure Sugar Level in Diabetes Using Saliva (Design Patent)	24.06.2025	Granted / Issued

SHORT-TERM COURSES / VALUE ADDED COURSES

Sl. No.	Course Name	Training Organization	Duration
1	AR-VR – Game Development	Unity Technology – South India's First Unity Centre of Excellence, St. Thomas International Centre, Chennai	7–12 Dec 2021 (06 Days)

Sl. No.	Course Name	Training Organization	Duration
2	Artificial Intelligence Certification course integrating Python, Machine Learning & Deep Learning	JPA Solution, Velachery, Chennai	Weekend (Sunday) six-month course, 10.02.2019 to 20.10.2019

DETAILS OF BOOK / CHAPTER PUBLISHED

Sl. No.	Title	Publication / ISBN or DOI
1	Heterogeneous Graph Convolution Cell Attention Network with Skill Optimization Based Jasmine Fragrance Detection — In: Modeling, Simulation and Optimization: Proceedings of CoMSO 2024	Springer, 2026 DOI: 10.1007/978-981-95-4161-4_28
2	Machine Learning and Applications – Chapter 3: Multi-task Cascaded Neural Network for face verification using Keras-face model	Dipti Press Pvt. Ltd, ISBN 978-93-86923-47-9, Feb 2020

AWARDS AND ACHIEVEMENTS

Sl. No.	Award Name	Organization	Year
1	Dr. C V Raman, Achievement Award	IET, UK Young Professional Section	2012

PROFESSIONAL SOCIETIES MEMBERSHIP

Sl. No.	Name of the Professional Society	ID No
1	The Institution of Engineers (India) (IEI), Type – MIE	M-1755751
2	The Institute of Electrical and Electronics Engineers (IEEE)	90779944
3	The Institution of Engineering and Technology (IET), UK	1100258975

ADDITIONAL RESPONSIBILITIES

At St. Joseph College of Engineering (2011–2023):

- ISO Certification College Level Coordinator & Internal Auditor, 2017–2023
- Anna University Internal Academic Auditor, 2022–2023
- NAAC Criteria 2 Coordinator
- Placement Officer, Jan 2020 – Jan 2022
- NBA Accreditation ECE Dept. Coordinator
- SJCE Professional Society Faculty Adviser for IEI, IET & IEEE

- AICTE & Anna University Accreditation ECE Dept. Coordinator (5 years)
- Subject Allocation & Dept. Time Table In-Charge (4 years)
- International Conference ICIECS 2018 & 2019 Organizing Committee Member
- Department Lab Coordinator & Department File In-Charge (5 years)
- Industrial Visit Coordinator, 2016 & 2017
- Symposium Coordinator, 2012 & 2016
- Executive Committee Member, IET UK Young Professional Section, Chennai
- Coordinator, Faculty Development Program, 2013 & 2017
- Class Coordinator for Second Year Students, 2013, 2014, 2015 & 2017
- Discipline Committee Member, St. Joseph College of Engineering, Chennai

SUBJECTS HANDLED

- Artificial Intelligence
- Deep Learning
- Digital Image Processing
- Digital Principles and System Design
- Digital Principles and Computer Organization
- Digital Electronics
- Microprocessor and Microcontroller
- Communication System
- Electronic Devices
- Circuit Analysis

PERSONAL DATA

Date of Birth	:	21-06-1987
Sex	:	Male
Marital Status	:	Married
Father's Name	:	Mr. M. Kanakam (B.D.O Rtd.)
Languages Known	:	English, Tamil, Malayalam
Nationality	:	Indian
Phone	:	09176009797
Present Address	:	
Permanent Address	:	Mavuninavilai Veedu, Nithiravilai (P.O) – 629154, Kanyakumari District

AFFIRMATION

I hereby declare that the particulars of information and facts stated herein above are true and complete to the best of my knowledge and belief.

[Dr. K. MICHAEL MAHESH]